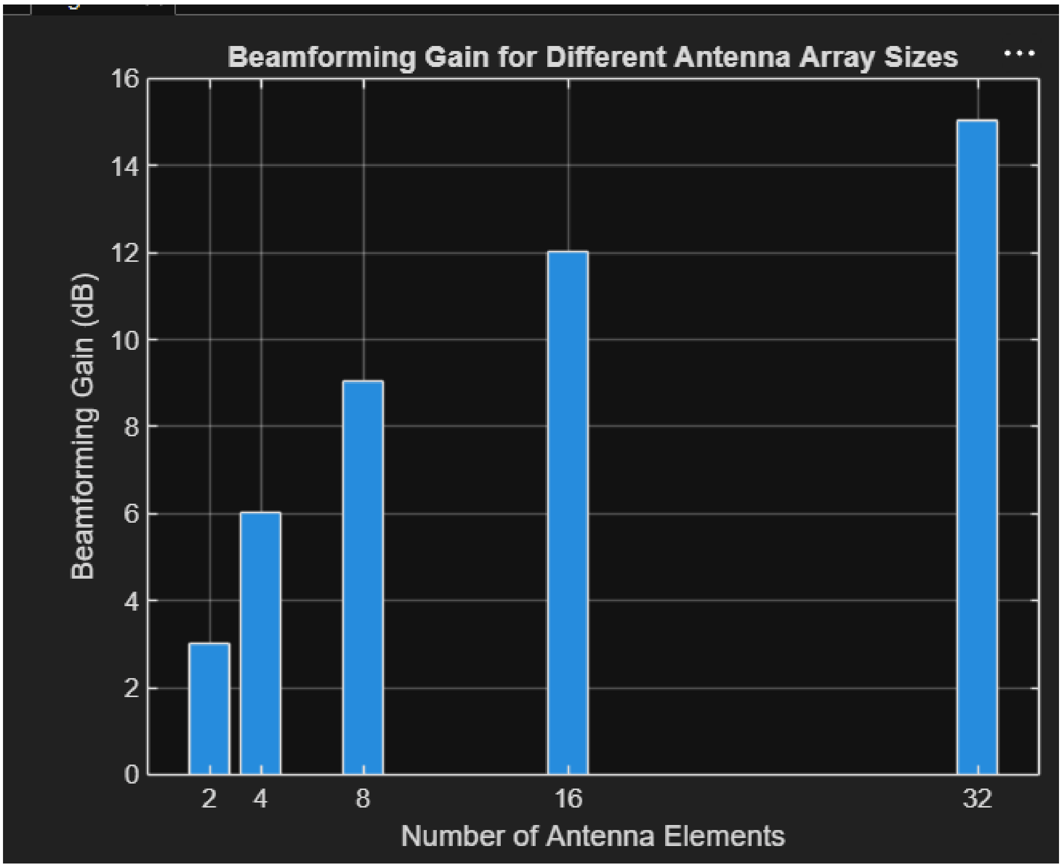
Experiment: 7

Analog Beamforming Analysis Using MATLAB

OBJECTIVE 1 :

CODE:

```
clc;
clear;
close all;
antennas = [2 4 8 16 32]; % Number of Antenna Elements
gain = 10*log10(antennas); % Beamforming Gain (dB)
figure
bar(antennas, gain)
grid on
xlabel('Number of Antenna Elements')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Antenna Array Sizes')
```



OBJECTIVE 2:

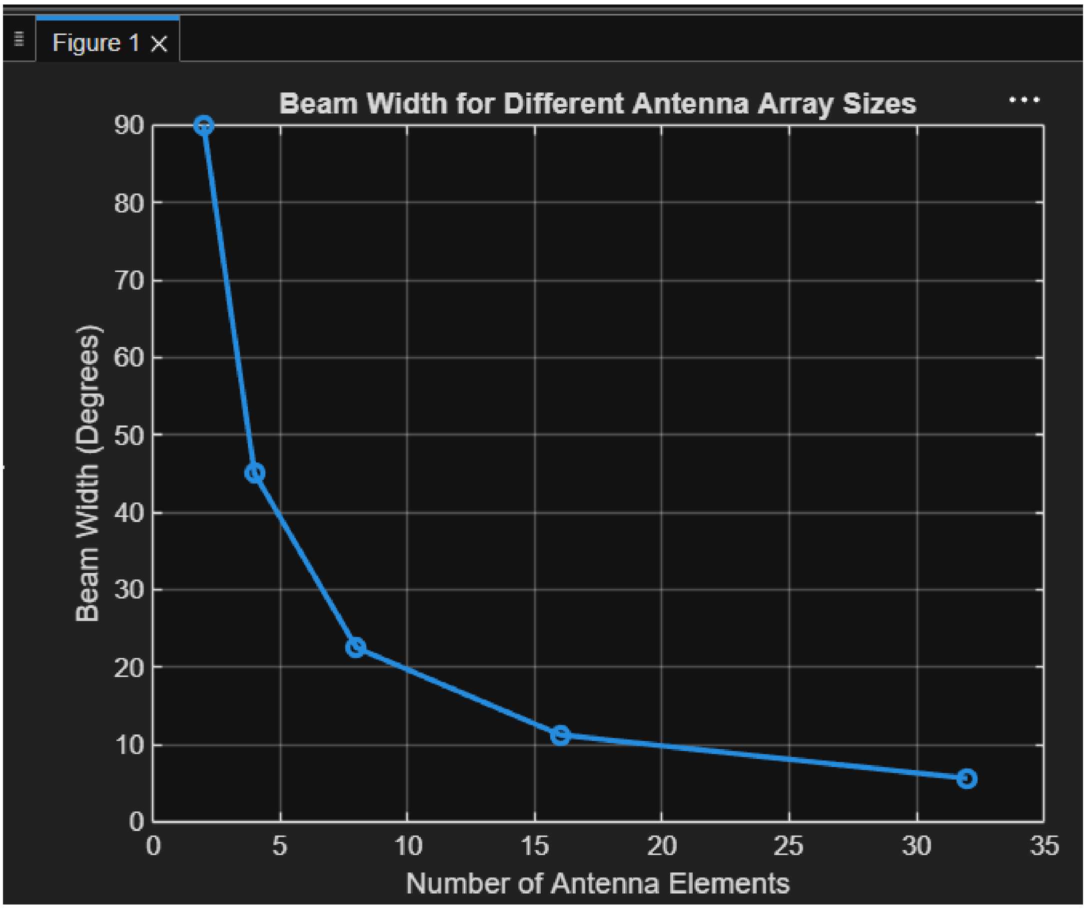
CODE:

```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width
(degrees)

figure

plot(antennas, beamwidth,'o-','LineWidth',2)

grid on
```

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')



OBJECTIVE 3:

CODE:

clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

gain = [1 1 1 1 1]; % Constant normalized gain

figure

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```